

Product Requirements Document (PRD)

Prototipe Sistem AI Document Detector (Versi Komprehensif)

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1. Pendahuluan

Dokumen ini mendefinisikan spesifikasi fungsional dan teknis secara lengkap untuk pengembangan prototipe web "AI Document Detector". Produk ini dirancang sebagai alat bantu komprehensif untuk menganalisis dan mendeteksi teks yang dihasilkan oleh kecerdasan buatan (Artificial Intelligence) dalam berbagai format dokumen. Sistem ini dirancang secara khusus untuk memenuhi kebutuhan akademik, seperti verifikasi tugas mahasiswa, makalah, dan integritas jurnal penelitian (seperti Jurnal Pendidikan Kimia).

2. Konsep & Fitur Utama Produk

2.1. Academic Integrity & Style Auditor

- **Per-Sentence Highlighting:** Menampilkan UI yang menyorot teks yang diunggah dengan warna (misalnya merah untuk buatan AI, hijau untuk buatan manusia, dan kuning untuk teks paraphrase AI).
- **Perplexity & Burstiness Score:** Membedah tingkat kerumitan kata (perplexity) dan variasi panjang kalimat (burstiness). Teks AI memiliki struktur yang statis dan seragam dibandingkan tulisan manusia yang dinamis.

2.2. Multi-Format & OCR Smart Verifier

- **Universal Upload:** Mendukung pengolahan langsung file berformat teks (.pdf dan .docx).
- **Optical Character Recognition (OCR):** Menggunakan engine (seperti Tesseract atau EasyOCR) untuk mengekstrak dan menganalisis teks yang bersumber dari gambar (.jpg, .png) atau dokumen hasil scan fisik.
- **Pendeteksi "Halusinasi" Format:** Memberi tanda jika struktur dokumen tidak wajar akibat kesalahan AI (misal: format kop surat yang salah).

2.3. Enterprise Metadata Forensics

- **Analisis Metadata:** Mengekstrak data tersembunyi di balik file (seperti waktu pembuatan, kapan terakhir diedit, dan author asli).
- **Time Anomaly Detection:** Mendeteksi ketidakwajaran, misalnya makalah 20 halaman memiliki Total Editing Time hanya 2 menit, memicu flag merah kemungkinan hasil copy-paste massal dari AI.

3. Alur Pemrosesan Data (Pipeline)

Sistem menggunakan arsitektur pemisahan antara Frontend (React) dan Backend (FastAPI). Urutan eksekusi pemrosesan data (pipeline) beroperasi sebagai berikut:

1. **Menerima File & Validasi Format:** Endpoint FastAPI menerima file. Sistem mengecek ekstensi. Jika gambar/scan, arahkan ke pipeline OCR. Jika teks murni, arahkan ke text parser.

2. **Ekstraksi Teks (Preprocessing):** Menggunakan pdfplumber atau EasyOCR untuk mengubah input menjadi string teks panjang (plain text) dan membersihkan karakter aneh (noise reduction).
3. **Segmentasi Kalimat:** Memecah teks utuh menjadi daftar kalimat menggunakan NLTK/SpaCy agar deteksi AI jauh lebih akurat dibanding membaca satu dokumen penuh.
4. **Inference (Skoring AI):** Mengirim daftar kalimat ke model Transformers. Dapatkan skor probabilitas per kalimat (0.0 = Manusia, 1.0 = AI) dan hitung rata-rata keseluruhan.
5. **Return JSON Response:** Dikirim kembali ke Frontend dalam bentuk JSON yang berisi: Skor Total, Status (Human/AI), dan array kalimat beserta warna highlight-nya.

4. Spesifikasi Teknologi (Tech Stack)

4.1. Core Framework

Lapisan	Teknologi Terpilih	Alasan Pemilihan
Frontend	Next.js (React) + Tailwind CSS	Ekosistem UI kaya, Tailwind mempercepat styling dashboard modern, menggunakan font Poppins, dan mendukung desain full responsive (mobile-first approach) yang optimal di perangkat desktop, tablet, maupun mobile.
Backend	Python (FastAPI)	Asynchronous, cepat menangani OCR/AI, auto-generate Swagger docs.
Database	PostgreSQL	Robust, relational database for data integrity and scalability.

4.2. Python Libraries (Lokal)

- **Ekstraksi/OCR:** pdfplumber / PyMuPDF (standar emas teks PDF), python-docx, EasyOCR (untuk scan gambar, support Bahasa Indonesia).
- **AI Detection Engine:** transformers (Hugging Face) menggunakan model seperti roberta-base-openai-detector. nltk/textstat untuk statistik.

4.3. Third-Party API (Alternatif / Beban Ringan)

- **GPTZero API:** Standar industri deteksi teks AI. Jika server lokal terbatas RAM, API ini mengembalikan probabilitas per kalimat secara akurat.
- **Google Cloud Vision API:** Alternatif jika EasyOCR lokal terlalu lambat; sangat cepat mengekstrak teks dari gambar bernoise.

5. Skema Database (ERD)

Tabel: users

- id (UUID / INTEGER) : Primary Key
- email (VARCHAR 100) : Unique
- name (VARCHAR 100) : Nama pengguna
- created_at (TIMESTAMP)

Tabel: scan_reports

- id (UUID / INTEGER) : Primary Key
- user_id (UUID / INTEGER) : Foreign Key (1-to-Many)
- file_name (VARCHAR 255)
- file_type (VARCHAR 10)
- word_count (INTEGER)
- ai_score (FLOAT)
- status_label (VARCHAR 20) : (Human, AI-Generated, Mixed)
- sentence_analysis (JSON / JSONB) : Array dari analisis tiap kalimat.
- created_at (TIMESTAMP)

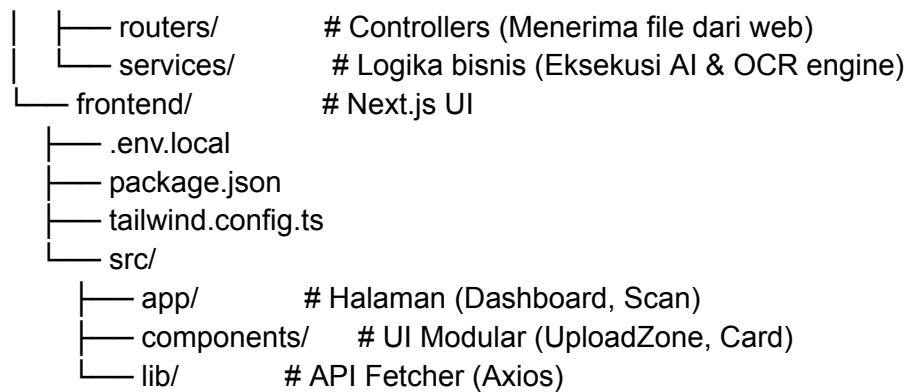
Keputusan Arsitektur Database

Penggunaan tipe data JSON / JSONB pada kolom `sentence\_analysis` merupakan strategi optimasi yang krusial. Alih-alih membuat tabel terpisah untuk menampung ribuan baris kalimat (yang akan memperlambat query), sistem hanya menyimpan 1 baris per dokumen. Frontend dapat langsung memparsing array JSON tersebut untuk merender warna per kalimat.

6. Struktur Repositori Proyek (Monorepo)

ai-doc-detector/

```
|— AGENTS.md           # Dokumentasi utama proyek & Tracker Tugas
|— README.md
|— backend/           # API FastAPI
|   |— .env           # Secret API Keys
|   |— requirements.txt
|   |— main.py         # Entry point FastAPI
|   |— models.py       # Skema database
|   |— schemas.py      # Pydantic validation
```



Alasan Efisiensi Struktur:

- **Separation of Concerns:** Pemisahan antara "routers" dan "services" pada backend. Jika mesin AI diganti, hanya folder services yang diubah tanpa merusak API.
- **Keamanan Environment:** Backend dan Frontend memiliki file .env terpisah untuk mencegah kebocoran API Key.
- **Kesesuaian Tipe Data:** Validasi Pydantic di Python dan TypeScript di React memastikan format JSON yang terkirim dan diterima konsisten, mencegah UI crash.

7. Metodologi Orkestrasi Proyek

Sebagai standar kerja, proyek ini mengadopsi pendekatan "Project Orchestrator". Setiap langkah analisis, perencanaan arsitektur, dan eksekusi akan dicatat ke dalam file "AGENTS.md" sebagai sumber kebenaran (source of truth) dokumentasi proyek. Sistem dikembangkan secara modular di mana backend API dibangun dan diuji terlebih dahulu sebelum UI dasbor diintegrasikan.

7. Design System Reference

None

version: alpha

name: Awwwards Mono Editorial

description: A restrained, high-contrast editorial system built around dense content, strong typography, and minimal chrome.

colors:

```
primary: "#222222"

secondary: "#6B7280"

tertiary: "#E5E7EB"

neutral: "#F8F8F8"

surface: "#FFFFFF"

on-surface: "#222222"

error: "#D92D20"

border: "#E5E7EB"

muted: "#D1D5DB"
```

typography:

```
headline-display: {fontFamily: "Inter Tight", fontSize:
"82px", fontWeight: 700, lineHeight: "98px", letterSpacing:
"0px"}
```

```
headline-lg: {fontFamily: "Inter Tight", fontSize: "54px",
fontWeight: 400, lineHeight: "65px", letterSpacing: "0px"}
```

```
headline-md: {fontFamily: "Inter Tight", fontSize: "35px",
fontWeight: 400, lineHeight: "65px", letterSpacing: "0px"}
```

```
headline-sm: {fontFamily: "Inter Tight", fontSize: "23px",
fontWeight: 400, lineHeight: "28px", letterSpacing: "0px"}
```

```
body-lg: {fontFamily: "Inter Tight", fontSize: "18px",
fontWeight: 400, lineHeight: "28px", letterSpacing: "0px"}
```

```
body-md: {fontFamily: "Inter Tight", fontSize: "15px",
fontWeight: 300, lineHeight: "28px", letterSpacing: "0px"}
```

```
body-sm: {fontFamily: "Inter Tight", fontSize: "14px",
fontWeight: 300, lineHeight: "24px", letterSpacing: "0px"}
```

```
label-lg: {fontFamily: "Inter Tight", fontSize: "18px",  
fontWeight: 400, lineHeight: "28px", letterSpacing: "0px"}
```

```
label-md: {fontFamily: "Inter Tight", fontSize: "15px",  
fontWeight: 400, lineHeight: "22px", letterSpacing: "0px"}
```

```
label-sm: {fontFamily: "Inter Tight", fontSize: "12px",  
fontWeight: 400, lineHeight: "16px", letterSpacing: "0px"}
```

```
caption: {fontFamily: "Inter Tight", fontSize: "12px",  
fontWeight: 300, lineHeight: "16px", letterSpacing: "0px"}
```

rounded:

```
none: 0px
```

```
sm: 4px
```

```
md: 8px
```

```
lg: 12px
```

```
x1: 16px
```

```
full: 9999px
```

spacing:

```
xs: 6px
```

```
sm: 14px
```

```
md: 24px
```

```
lg: 32px
```

```
x1: 48px
```

```
gutter: 24px
```

```
section: 48px
```

components:

```
button-primary: {backgroundColor: "transparent", textColor:
"{colors.surface}", typography: "{typography.body-lg}",
rounded: "{rounded.md}", padding: "27px 29px", size: "72px"}
```

```
button-primary-hover: {backgroundColor: "{colors.primary}",
textColor: "{colors.surface}", typography:
"{typography.body-lg}", rounded: "{rounded.md}", padding:
"27px 29px", size: "72px"}
```

```
button-secondary: {backgroundColor: "transparent",
textColor: "{colors.surface}", typography:
"{typography.body-lg}", rounded: "{rounded.md}", padding:
"27px 29px", size: "72px"}
```

```
button-link: {backgroundColor: "transparent", textColor:
"{colors.on-surface}", typography: "{typography.body-lg}",
rounded: "{rounded.none}", padding: "0px"}
```

```
card: {backgroundColor: "{colors.neutral}", textColor:
"{colors.on-surface}", rounded: "{rounded.md}", padding:
"16px"}
```

```
input: {backgroundColor: "{colors.tertiary}", textColor:
"{colors.on-surface}", typography: "{typography.body-md}",
rounded: "{rounded.md}", padding: "14px"}
```

```
chip: {backgroundColor: "{colors.primary}", textColor:
"{colors.surface}", typography: "{typography.label-sm}",
rounded: "{rounded.full}", padding: "6px"}
```

Awwwards Mono Editorial

Overview

This interface feels like a curated editorial showcase: minimalist, confident, and heavily typographic. It is designed for design-conscious professionals who expect fast scanning, strong hierarchy, and a premium but understated visual tone. The layout is spacious and content-led, with a quiet light background that lets black-and-white imagery and oversized type take center stage.

Colors

- **Primary (#222222):** A deep graphite used for the main text, key UI chrome, and dark fills. It provides the system's crisp, high-contrast backbone.
- **Secondary (#6B7280):** A muted neutral for supporting text and less prominent metadata. Use it sparingly so the hierarchy stays sharp.
- **Tertiary (#E5E7EB):** A soft divider and control tint used for borders, inputs, and subtle surfaces.
- **Neutral (#F8F8F8):** The site's light canvas color, creating a clean gallery-like environment that keeps focus on content.
- **Surface (FFFFFF):** Reserved for floating cards, buttons, and high-contrast UI elements that need to read as lifted from the page.
- **On-surface (#222222):** The default text color on light surfaces; it matches the primary graphite for consistency.
- **Border (#E5E7EB):** A restrained border tone for cards, pills, and form fields when a visible edge is needed.
- **Muted (#D1D5DB):** A softer neutral for inactive states and low-emphasis UI details.

Typography

The system is built on Inter Tight, which gives the UI a compact, modern, editorial feel. Headlines are large, clean, and mostly weight 400 except for the display treatment, which uses 700 for maximum impact. Body text stays small and airy, using 300–400 weight with generous line height to balance the dense visual content. Use headline-display for hero moments and major page statements, headline-lg through headline-sm for section hierarchy, and body-md for most supporting copy. Labels and navigational text should rely on label-md and label-sm, keeping letter spacing neutral and avoiding decorative uppercase treatments unless a specific component calls for it. The overall typographic rhythm is restrained rather than flashy, with weight and size doing most of the work.

Layout

The composition favors a fixed-max-width editorial grid with large outer margins and generous vertical breathing room. Navigation is compressed into a single top bar, while hero content sits centered and isolated to emphasize importance. Section spacing should follow the 6/14/24/32/48px scale, with 24px and 48px doing most of the heavy lifting for content separation. Cards and panels should use ample internal padding, especially in featured modules and toolbars, but avoid overly dense stacking. The page feels spacious even when content is visually busy because whitespace is used strategically to isolate the biggest text and imagery.

Elevation & Depth

Depth is handled mostly through contrast, layering, and borders rather than shadows. Surfaces are flat and calm, with subtle bordered cards and darker overlays providing separation where necessary. The main elevated moments come from tonal contrast: dark panels on light canvas, light buttons on dark panels, and floating cookie or tool trays that read as secondary layers.

Shapes

The shape language is softly squared and practical. Small radii around 8px are the dominant look, giving buttons, cards, inputs, and chips a polished but not playful finish. Full pills are used selectively for compact tags and badges, while the rest of the system stays close to rectangular for an architectural editorial feel.

Components

Buttons should feel compact, bordered, and typography-forward. `button-primary` and `button-secondary` are visually similar in this system: transparent backgrounds, white text on dark contexts, 8px corners, and generous padding around a 72px minimum touch target. They should remain calm and precise rather than chunky; hover states can invert into subtle fills or stronger contrast without adding shadow. `button-link` is plain text with no border or background and is best used in the top nav or inline actions. Cards should use card styling: light background, 1px border, 8px radius, and 16px padding. They should hold content quietly, without shadows or loud accent colors. Inputs should match the same restrained vocabulary, using a pale tinted background, visible but soft borders, and body-sized text. Chips and pills should use chip with full rounding and compact padding, especially for status tags and navigation filters. Lists and metadata rows should stay tight and horizontally aligned, with typography doing the emphasis work. Any icon buttons or compact controls should mirror the same 8px radius and subdued border logic. Avoid ornate component states; the system works best when interactions remain minimal and editorial.

Do's and Don'ts

- Do keep layouts spacious and centered when showcasing hero content or featured work.
- Do use Inter Tight consistently so typography remains the primary visual signature.
- Do rely on contrast, borders, and layering instead of heavy shadows.
- Do keep radii modest, with 8px as the default for most controls and cards.
- Don't introduce bright accent colors that compete with the monochrome palette.
- Don't add decorative gradients, glows, or glossy effects.
- Don't make buttons or inputs overly rounded or overly filled; preserve the restrained, premium feel.

- Don't crowd the page with dense blocks of equal-weight text; maintain a clear hierarchy.